Membership Inference as a Common Yardstick: Evaluating Five Families of Privacy-Preserving Defences for Healthcare Machine Learning

Anjali Ashtankar
Roll No. 16014223012, Batch B1 - Security and Privacy for Machine Learning (IA-2 manuscript draft)
Code and data: https://github.com/anjaliash005/MLCS-IA

***Abstract*—Membership inference attacks (MIAs) reveal whether a record was used to train a model, which in healthcare can disclose a diagnosis. We use a black-box MIA as a common yardstick to compare five families of privacy-preserving techniques on a diabetes-risk task: regularisation, central differential privacy (DP-GD with a zCDP accountant), local differential privacy (Piecewise mechanism and LDP Naive Bayes), privacy-preserving data mining (Mondrian k-anonymity, l-diversity, input noise, output restriction) and compressive privacy (PCA/DCA/DUCA). Across 10 paired splits with Kerckhoffs-style shadow models, an overfit MLP leaks membership with attack AUC 0.77. DP-GD at ε = 4 keeps 74.7% accuracy at AUC 0.51, while label-only outputs and k-anonymity reduce AUC only to 0.67 and 0.60. A single DCA component keeps 77.2% accuracy while reducing age inference from 1.00 to 0.62 balanced accuracy. The results replicate on a second medical dataset.**

***Index Terms*—membership inference, differential privacy, local differential privacy, k-anonymity, compressive privacy, healthcare machine learning.**

I. INTRODUCTION

Releasing a trained model is not the same as releasing nothing about its training data. Shokri et al. [1] showed that black-box outputs reveal training-set membership, and Yeom et al. [2] linked this leakage to overfitting. Many defences have been proposed, but they come from separate communities (differential privacy [5], local DP [10], data anonymisation [14], compressive privacy [19]; see [21] for a unified treatment) and are usually evaluated with incompatible metrics. A practitioner choosing among them for a clinical model has no common basis for comparison.

Contributions. (1) A reproducible harness in which every defence is a training recipe attacked by shadow models trained with the *same* recipe. (2) First-principles implementations of DP-GD with zCDP accounting, four LDP mechanisms with OUE-based LDP Naive Bayes, Mondrian k-anonymity with l-diversity, and DCA/DUCA projections. (3) A paired, multi-seed comparison on two public medical datasets, including an ablation that separates the effect of DP noise from that of shortened training, and a comparison of measured advantage with the theoretical DP bound.

II. BACKGROUND AND RELATED WORK

Membership inference. Shadow-model attacks [1] train a meta-classifier on outputs of models with known membership; loss-threshold attacks [2] need no shadows; label-only attacks [3] work even when scores are hidden; LiRA [4] targets the low-FPR regime. *Differential privacy.* DP-SGD [6] clips per-example gradients and adds Gaussian noise; zCDP [7] composes Gaussian steps tightly; a mechanism that is (ε, δ)-DP limits any membership test to TPR − FPR ≤ (e^ε − 1 + 2δ)/(e^ε + 1) [8]. *Local DP* removes trust in the collector: Duchi's mechanism [10], the Piecewise and Hybrid mechanisms [11], and frequency oracles such as OUE [12] that enable LDP Naive Bayes [13]. *Syntactic anonymity*: k-anonymity [14] via Mondrian [15], l-diversity [16] and t-closeness [17]; random input perturbation [18]. *Compressive privacy* [19], [20] releases low-dimensional projections designed to keep a utility task and suppress a privacy task.

III. THREAT MODEL AND METHOD

A hospital trains a binary classifier on n = 192 records (members); another 192 are non-members and serve as the test set. The attacker holds 384 records from the same distribution, knows the training recipe and any published transform, and queries the model as a black box. The API returns full probabilities, probabilities rounded to 0.1, or labels only. For each configuration the attacker trains 8 shadow models with the defended recipe, extracts features (true-class probability, loss, entropy, maximum probability, correctness, logit, label), and fits a logistic-regression attack. A loss-threshold attack calibrated on the shadows is run alongside it. We report attack ROC-AUC, calibrated advantage and maximum advantage.

Defences. **M1:** logistic regression, MLP-32, MLP-128x2 without regularisation (baseline), the same with L2 = 3 and early stopping, and a deep random forest. **M2:** full-batch DP-GD (T = 200, C = 1, $\delta = 10^{-5}$), where $\varepsilon = \rho + 2\sqrt{(\rho \ln(1/\delta))}$ and $\rho = T/(2\sigma^2)$; σ = 18.3 gives ε = 4. **M3:** each patient perturbs 10 features and the label with ε/11 each (Piecewise + randomised response), or reports one sampled (class, bin) pair through OUE for LDP-NB. **M4:** Mondrian on Age, Pregnancies and BMI with $k \in \{5, 10, 25\}$ and optional l = 2; Gaussian input noise; output rounding and label-only APIs. **M5:** PCA, DCA ($S_B + \rho' I$ vs $S_W + S_B + \rho I$) and DUCA (adds γS_B of the privacy label, Age >= 35, to the denominator). An unsupervised DP k-means demonstration [9] is reported in the companion technical report.

Data. Pima Indians Diabetes (768 records, 268 positive) [22] and Breast Cancer Wisconsin Diagnostic (569 records) [23]. Code, data and results are available at https://github.com/anjaliash005/MLCS-IA. Everything is implemented in NumPy, with scikit-learn [24] used for the

non-private models and metrics. All scaling uses fixed public bounds, so preprocessing consumes no privacy budget. Results are means over seeds 0-9; paired Wilcoxon tests compare each configuration with the baseline.

IV. RESULTS

TABLE I. *Pima results (mean of 10 seeds). AUC* = worst-case attack AUC; p = Wilcoxon vs baseline, shadow AUC.*

Configuration	Acc	AUC*	p (AUC)
Baseline MLP-128x2	0.719	0.765	ref.
Logistic regression	0.773	0.511	0.002
+L2 + early stop	0.711	0.505	0.002
Deep RF	0.759	0.798	0.010
Baseline, label-only API	0.719	0.666	0.002
DP-GD MLP ε=1	0.594	0.505	0.002
DP-GD MLP ε=4	0.747	0.511	0.002
DP-GD LR ε=4	0.758	0.516	0.002
LDP-NB ε=4	0.668	0.514	0.002
Mondrian k=25	0.694	0.598	0.002
Mondrian k=10, l=2	0.714	0.640	0.002
Input noise σ=0.5	0.689	0.521	0.002
DCA k=1	0.745	0.537	0.002
DUCA k=2	0.695	0.669	0.002

Overfitting (M1). The baseline reaches train-test gap 0.28 and attack AUC 0.77; the deep random forest leaks even more (0.80) despite higher accuracy. Regularisation lowers AUC to 0.49 without a significant accuracy change, and plain logistic regression is both the most accurate (77.3%) and essentially leak-free (0.51).

Central DP (M2). Accuracy rises with ε, from 59.4% at ε = 1 to 74.7% at ε = 4, while attack AUC stays near 0.5. The ablation shows that the same network trained without noise for 2000 steps leaks (AUC 0.74), but for 200 steps it does not (0.52). Measured maximum advantage stays within the random-score noise floor and far below the bound of [8].

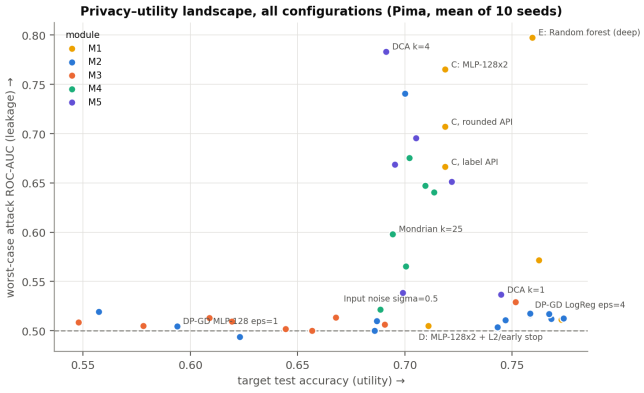


Fig. 1. Privacy-utility landscape of all configurations on Pima (lower right is better).

Local DP (M3). All mechanisms match their theoretical variance; Piecewise beats Duchi from ε = 1 upward, as in [11]. LDP-trained models are near chance for MIA, but utility is data-limited: LDP-NB reaches 66.8% at ε = 4 (non-private 75.2%). A bootstrap simulation with 50,000 users gives 74.1% at ε = 1.

PPDM (M4). Label-only outputs leave AUC at 0.67 because an overfit model is right far more often on its members [3]. k-anonymity with k = 25 gives AUC 0.60. Without l-diversity, 15.1% of records at k = 10 fall in diagnosis-homogeneous classes; l = 2 removes this exposure for a small increase in NCP (0.13 to 0.14).

Compressive privacy (M5). PCA retains reconstructable, age-revealing variance (age balanced accuracy 0.80 at k = 1). DCA concentrates the diagnostic signal: one component yields 77.2% accuracy (raw 77.3%), age inference 0.62 and reconstruction NRMSE 0.96 at k = 2. DUCA lowers age inference to 0.51. Models trained on z can still memorise once z has enough dimensions (DCA k >= 2, PCA k >= 4), so CP protects released data rather than models.

TABLE II. *Compressive privacy on Pima: utility (logistic regression on z), attribute-inference attacker (Age >= 35) and linear reconstruction of x.*

Projection	Utility acc	Age bal. acc	Recon. NRMSE
Raw features	0.773	1.000	0.000
PCA, k = 1	0.651	0.803	0.617
PCA, k = 4	0.758	0.902	0.333
DCA, k = 1	0.772	0.623	0.957
DCA, k = 2	0.771	0.653	0.956
DUCA, k = 1	0.767	0.512	0.968
DUCA, k = 2	0.767	0.536	0.967

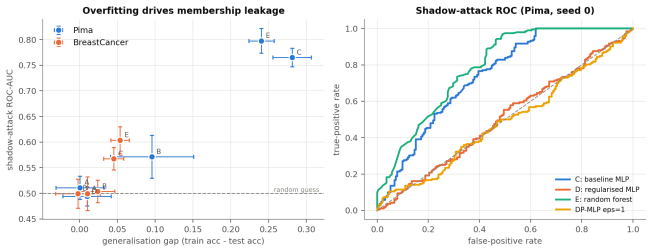


Fig. 2. Left: attack AUC vs generalisation gap (models A-E, both datasets). Right: shadow-attack ROC curves on one Pima split.

Replication. On WDBC the baseline generalises well (accuracy 95.5%, AUC 0.57), while the deep random forest remains leaky (0.60). The ordering of defences is unchanged.

V. DISCUSSION AND LIMITATIONS

On small tabular data, matching model capacity to the data is the cheapest defence. Beyond that, formal training-side protection is the most robust choice for model release. Compressive projections are the most effective for data release. Syntactic and output-side measures should be complements, not replacements. Our study is limited to two small tabular datasets, to average-case attackers (AUC rather than TPR at low FPR [4]) and to full-batch DP-GD without subsampling amplification, so the reported ε is conservative. Hyperparameters of the private models were not selected privately.

Ethics. Only public, de-identified data and locally trained models were used. The Pima data originate from an Indigenous community and their benchmark use has been criticised, so no claims about the community are made and a second dataset is included.

VI. CONCLUSION

Using one attack as the yardstick turns a collection of privacy techniques into comparable options. In our healthcare setting,

moderate-ε DP-GD offers the best guarantee-utility balance for releasing models, DCA/DUCA offer the best balance for releasing data, and LDP becomes attractive once user bases reach tens of thousands. Future work will add subsampled DP-SGD with RDP accounting, LiRA-style attacks and combined DCA + DP pipelines on larger clinical datasets.

REFERENCES

[1] R. Shokri, M. Stronati, C. Song and V. Shmatikov, "Membership inference attacks against machine learning models," in *Proc. IEEE Symp. Security and Privacy*, 2017, pp. 3-18.

[2] S. Yeom, I. Giacomelli, M. Fredrikson and S. Jha, "Privacy risk in machine learning: Analyzing the connection to overfitting," in *Proc. IEEE CSF*, 2018, pp. 268-282.

[3] C. A. Choquette-Choo, F. Tramèr, N. Carlini and N. Papernot, "Label-only membership inference attacks," in *Proc. ICML*, 2021.

[4] N. Carlini, S. Chien, M. Nasr, S. Song, A. Terzis and F. Tramèr, "Membership inference attacks from first principles," in *Proc. IEEE Symp. Security and Privacy*, 2022.

[5] C. Dwork and A. Roth, "The algorithmic foundations of differential privacy," *Foundations and Trends in Theoretical Computer Science*, vol. 9, no. 3-4, 2014.

[6] M. Abadi, A. Chu, I. Goodfellow, H. B. McMahan, I. Mironov, K. Talwar and L. Zhang, "Deep learning with differential privacy," in *Proc. ACM CCS*, 2016, pp. 308-318.

[7] M. Bun and T. Steinke, "Concentrated differential privacy: Simplifications, extensions, and lower bounds," in *Proc. TCC*, 2016, pp. 635-658.

[8] P. Kairouz, S. Oh and P. Viswanath, "The composition theorem for differential privacy," in *Proc. ICML*, 2015.

[9] D. Su, J. Cao, N. Li, E. Bertino and H. Jin, "Differentially private k-means clustering," in *Proc. ACM CODASPY*, 2016.

[10] J. C. Duchi, M. I. Jordan and M. J. Wainwright, "Local privacy and statistical minimax rates," in *Proc. IEEE FOCS*, 2013.

[11] N. Wang, X. Xiao, Y. Yang, J. Zhao, S. C. Hui, H. Shin, J. Shin and G. Yu, "Collecting and analyzing multidimensional data with local differential privacy," in *Proc. IEEE ICDE*, 2019.

[12] T. Wang, J. Blocki, N. Li and S. Jha, "Locally differentially private protocols for frequency estimation," in *Proc. USENIX Security*, 2017.

[13] E. Yilmaz, M. Al-Rubaie and J. M. Chang, "Locally differentially private naive Bayes classification," arXiv:1905.01039, 2019.

[14] L. Sweeney, "k-anonymity: A model for protecting privacy," *Int. J. Uncertainty, Fuzziness and Knowledge-Based Systems*, vol. 10, no. 5, 2002.

[15] K. LeFevre, D. J. DeWitt and R. Ramakrishnan, "Mondrian multidimensional k-anonymity," in *Proc. IEEE ICDE*, 2006.

[16] A. Machanavajjhala, D. Kifer, J. Gehrke and M. Venkitasubramaniam, "l-diversity: Privacy beyond k-anonymity," *ACM TKDD*, vol. 1, no. 1, 2007.

[17] N. Li, T. Li and S. Venkatasubramanian, "t-closeness: Privacy beyond k-anonymity and l-diversity," in *Proc. IEEE ICDE*, 2007.

[18] R. Agrawal and R. Srikant, "Privacy-preserving data mining," in *Proc. ACM SIGMOD*, 2000.

[19] S. Y. Kung, "Compressive privacy: From information/estimation theory to machine learning," *IEEE Signal Processing Magazine*, vol. 34, no. 1, pp. 94-112, 2017.

[20] S. Y. Kung, T. Chanyaswad, J. M. Chang and P. Wu, "Collaborative PCA/DCA learning methods for compressive privacy," *ACM Trans. Embedded Computing Systems*, vol. 16, no. 3, 2017.

[21] J. M. Chang, D. Zhuang and G. D. Samaraweera, *Privacy-Preserving Machine Learning*. Manning, 2023.

[22] J. W. Smith, J. E. Everhart, W. C. Dickson, W. C. Knowler and R. S. Johannes, "Using the ADAP learning algorithm to forecast the onset of diabetes mellitus," in *Proc. Symp. Computer Applications in Medical Care*, 1988, pp. 261-265.

[23] W. N. Street, W. H. Wolberg and O. L. Mangasarian, "Nuclear feature extraction for breast tumor diagnosis," in *Proc. SPIE Biomedical Image Processing and Biomedical Visualization*, vol. 1905, 1993.

[24] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *JMLR*, vol. 12, pp. 2825-2830, 2011.